Vivian Li

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EDUCATION

Brown University | Providence, RI, USA

Sc.M. in Electrical and Computer Engineering (Research Track) September 2025 – May 2026

- Recipient of School of Engineering Master's tuition award

Sc.B. in Computer Science (Visual Computing + Design) September 2021 – May 2025

- Cross-registration at **Massachusetts Institute of Technology (MIT)** and **Rhode Island School of Design (RISD)**

EXPERIENCE

Brown University – Research Assistant, Computational Design Group Sept. 2025 – *Present*

- First-author work on an end-to-end computational design and fabrication system for programmable surface deformation in embroidered textiles (UIST 2026 Short Paper, accepted).
- Co-authored DicePlay (SIGGRAPH 2026), a grammar-based optimization framework for a modular display made of dice.
- Ongoing research on mixed discrete–continuous optimization for multi-objective garment design.
- Advised by **Prof. Adriana Schulz.**

Walt Disney Animation Studios – Production Technology Intern | Burbank, CA, USA June 2024 – August 2024

- Designed, implemented, and deployed a C++/Qt UI State Debugger in Presto (Pixar's proprietary animation & rigging software) to visualize complex debugging states and support artist and production workflows.
- First intern project adopted into production pipelines at both Pixar and Walt Disney Animation Studios.

MIT – Research Assistant, Fibers@MIT | Cambridge, MA, USA January 2024 – June 2024

- Developed a knit-integration technique for embedding fiber-based computation in textiles, enabling fabric-scale sensing and distributed inference; co-authored publication in *Nature* (2025).
- Advised by **Prof. Yoel Fink.**

Tsinghua University – Research Intern, Future Lab | Beijing, China June 2023 – August 2023

- Led research on a generative pipeline for creating buildable LEGO models and assembly instructions from text and images, investigating 3D generation and LEGO representations.
- Supervised by **Prof. Yingqing Xu**.

Microsoft Research Asia – Internet Graphics Intern | Beijing, China July 2022 – August 2022

- Researched lighting and virtual environment design for *Project VirtualCube*, an immersive telepresence system that renders remote participants in real time while preserving 3D geometry, texture, and mutual eye gaze.
- Supervised by **Dr. Xin Tong.**

Brown University – Undergraduate Research Assistant, Dash | Providence, RI, USA Jan. 2022 – Jan. 2023

- Developed a full-stack multicolumn notetaking feature for Dash, a collaborative hypermedia web application supporting scalable document-based knowledge workflows, with document linking and markup.
- Advised by **Prof. Andries van Dam.**

PUBLICATIONS

StitchSwitch: Programmable Surface Deformation and Bistability in Embroidered Textiles
ACM UIST 2026 (Conditionally Accepted)
Vivian Li, Milin Kodnongbua, Heather Robertson, Yiyue Luo, Adriana Schulz

DicePlay: A Modular Canvas for Physical Image Composition
ACM SIGGRAPH 2026 (To Appear)
Milin Kodnongbua*, Zihan Jack Zhang*, Shishi Xiao*, **Vivian Li**, Heather Robertson, Rulin Chen, David Laidlaw, Adriana Schulz

A Single-fibre Computer Enables Textile Networks and Distributed Inference
Nature 2025
Nikhil Gupta, Henry Cheung, Syamantak Payra, Gabriel Loke, Jenny Li, Yongyi Zhao, Latika Balachander, Ella Son, **Vivian Li**, Samuel Kravitz, Sehar Lohawala, John Joannopoulos, and Yoel Fink

SELECTED PROJECTS

Kinetic Pixels | Robotic Swarm Simulation October – December 2025

- Designed a robotic swarm simulation for rendering dynamic physical mosaics using cellular automata and collision-aware path planning.

Aligned | Embedded Electronics Design October 2024 – April 2025

- Built a posture-sensing system for textile integration with a custom PCB, I²C accelerometers, haptic feedback, and real-time machine learning for posture classification.

Color Me Noisy | Advanced Computer Graphics Project March – May 2025

- Reimplemented an example-based rendering pipeline that stylizes 3D animations as temporally coherent hand-colored artwork using multi-scale texture synthesis.

Spirit Oasis | Computer Graphics Project November – December 2024

- Recreated the Spirit Oasis scene from *Avatar: The Last Airbender*, implementing procedural animation, custom mesh deformers, and real-time rendering in Three.js with assets modeled in Blender.

Aerial Reverie | Coordinated Robotics VR Short Film March – May 2024

- Programmed and directed an immersive VR storyworld featuring a dynamic cityscape built by coordinated Crazyflie drone flight paths using Python and ROS.

PawLink | Computational Textiles November – December 2023

- Designed and developed a knitted smart collar integrating fiber-based sensing and on-device machine learning for real-time animal health monitoring.

SKILLS

Languages: C++, Python, JavaScript/TypeScript, Java, HTML/CSS, MATLAB, LaTeX
Software & Frameworks: Qt, React, OpenGL, Arduino, Linux
3D/2D Design: Blender, Maya, Unity, Fusion 360, Marvelous Designer, Adobe Creative Suite, Figma
Fabrication: PCB & Schematic Design, 3D Printing, Laser Cutting, Knitting, Sewing, Embroidery

TEACHING EXPERIENCE

Head Teaching Assistant, CSCI1953B Computational Design and Fabrication November 2025 – May 2026

LEADERSHIP & SERVICE

Co-Founder & Web Lead, SenseScape: *Beyond Boundaries* | NYC, NY, USA October 2023 – May 2025

- Co-organized an innovation conference and exhibition at Microsoft Garage NYC featuring XR installations, live demos, and panel discussions for 200+ attendees. Led development of an immersive 3D exhibition website archiving conference works with future VR accessibility in mind.

Design Lead, Hack@Brown | Providence, RI, USA September 2023 – March 2025

- Led the design team in developing the website and 3D visual identity for Hack@Brown 2025 and 2024, attracting 500+ participants from across the country each year.

Outreach Lead, Brown/RISD XR Hub | Providence, RI, USA September 2023 – December 2024

- Founding executive board member who led outreach to industry speakers and partners while organizing interdisciplinary XR events connecting students across Brown and RISD.

MEDIA

RISD News — "*RISD and Brown Students Present Innovation Conference at Microsoft Garage NYC*" (2025)
Brown Daily Herald — "*MIT, RISD and Brown researchers develop computers you can wear*" (2025)